

A Scalable Tracking System Using Sparse Dot-Based Markers

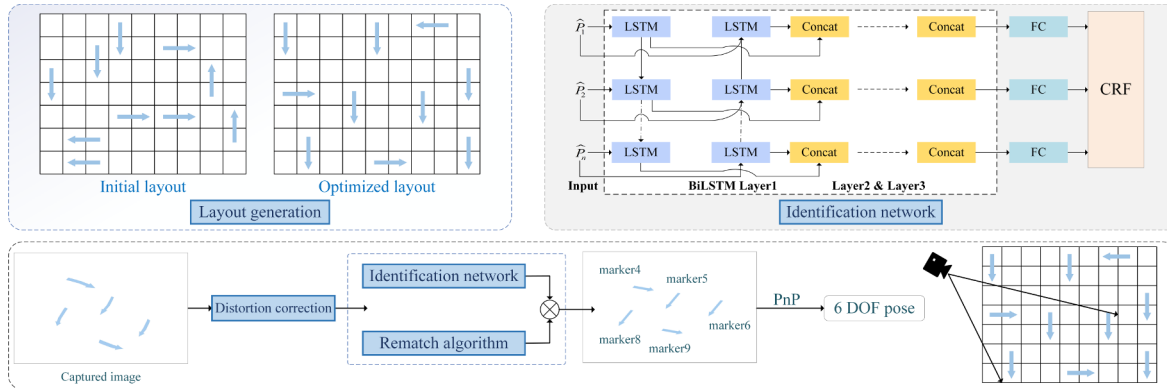
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Figure 1: Overview of the proposed method. Marker layout can be generated under given conditions, and we combine identification network with a rematching algorithm to achieve marker identification.

ABSTRACT

Localization in large-scale environment is fundamental for high-degree-of-freedom augmented reality (AR) and virtual reality (VR) applications. However, achieving stable and cost-effective wide-area tracking remains challenging, as existing methods often increase system cost and complexity when scaled to large spaces. This paper proposes a vision-based tracking system that encodes spatial relationships between markers to generate a planar layout. We propose a BiLSTM-based network for efficient marker recognition and introduce a robust rematching algorithm to enhance recognition accuracy. The proposed approach scales effectively with sparse markers while maintaining a high recognition rate.

Index Terms: Tracking, fiducial marker, layout generation, dots identification.

1 INTRODUCTION

Real-time, precise tracking in large-scale VR/AR scenes is crucial for enhancing user immersion and is considered one of the key development directions in expansive environments. Due to the simplicity of setup and high precision, "inside-out" tracking methods based on artificial marker recognition are better suited for the low-cost scaling of tracking scenarios, prompting a surge in related research. For example, STag [2] utilizes 48 circular bits in a square for better estimation accuracy. TopoTag [3] achieves high robustness using internal info, enabling full-bit ID encoding without shape constraints.

Most research has focused on improving the quality of an individual marker, while these markers can be easily deployed using black-and-white prints, the number increases significantly as the tracking

area expands, and the contradiction between tracking accuracy and robustness against occlusion is rather difficult to resolve [1]. To address this, this paper proposes a tracking system based on sparse marker recognition. As illustrated in Fig. 1, an optimized layout can be generated by encoding the spatial relationships among markers, and we achieve fast and accurate marker recognition by employing deep learning method with a rematching algorithm.

2 METHOD

Our system framework consists of two components: the layout generation module and the marker recognition module.

2.1 Layout Generation Module

In our approach, we construct an individual marker from dots with fixed relative positions, and all markers in the layout share the same internal structure. While the number of dots is not fixed, each constructed marker must exhibit sufficient spatial distinguishability. This means that, under normal conditions (excluding severe tilt or distortion), the spatial distribution of the dots in a single marker should allow for unambiguous identification of each dot, requiring $n > 2$. With the tracking space defined, we select either the floor or ceiling plane for marker placement and discretize the plane into a grid. Each marker is placed at a fixed position in two adjacent grid cells and can be represented as a directional arrow. The layout is then formed by arranging these markers within the grid. Fig. 2 (a) shows four possible designs of an individual marker.

Positional features are defined as the relative distribution of markers within a certain plane distance. Since the position of a single marker within its grid is relatively fixed, we can identify virtual corner points through the known vertical relationships between them. By applying re-projection and coordinate scaling transformations to the regions defined by these virtual corner points, the relative positional features can be recovered, thus facilitating marker recognition. Ultimately, the 6 DOF pose can be determined using the PnP method.

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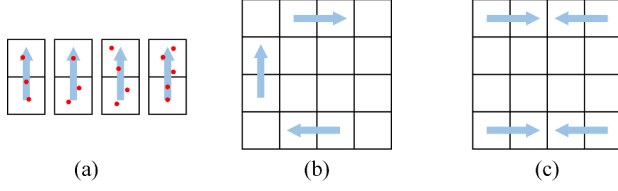


Figure 2: (a) four possible designs of the marker, (b) and (c) are examples of positional features.

The positional features of two markers U_1 and U_2 can be denoted as $F_{12} = \delta U_1 U_2$ and $F_{21} = \delta U_2 U_1$. It is important to note that if the two markers are centrally symmetric, it becomes impossible to determine the pose. With an additional marker U_3 , the positional features between each pair of markers are $F_{pair} = \{F_{12}, F_{13}, F_{21}, F_{23}, F_{31}, F_{32}\}$. Ideally, the layout is designed to prevent any duplicated positional features, but to achieve sufficient local marker density may inevitably lead to repeated features. In this context, we should notice that the three markers exhibit positional features as $F_{123} = \delta U_1 U_{23}$, $F_{213} = \delta U_2 U_{13}$, $F_{312} = \delta U_3 U_{12}$, and they are inherently distinct. Thus a unique pose can still be determined when all three markers are fully visible. As shown in Fig. 2 (b) only has features among the three markers, and (c) does not exhibit any features. As the number of markers increases, we only need to exclude repetitive features, while the remaining features are distinct enough to complete marker recognition.

The layout quality can be intuitively reflected by the feature distribution within the camera's field of view under different poses. To reduce computational complexity, pose and the view of camera are simplified on the 2D plane. Starting from an initial layout, we sequentially move each marker within its surrounding grid. For each updated layout, valid features are calculated first, then the layout quality is determined based on local marker density and feature distribution within the views. We iterate this process and introduce perturbations (random larger movements) to escape local optima, obtaining the final optimized layout.

2.2 Marker Recognition Module

Traditional algorithms for marker recognition struggle to robustly recognize dots of incomplete markers and face a significant increase in computational complexity as the number of dots grows. We thus employ deep learning, similar to existing research [4], for marker recognition under challenging conditions.

We formulate dots recognition as a classification task, where each dot in the layout is assigned a unique identifier. The network input are the 2D coordinate values of the dot to each of the other $N - 1$ detected dots: $(d_{x_i}, d_{y_i})_{i \in [1, N]}$, sorted in ascending order based on their distances from the reference dot. Considering the interrelation of all samples within a frame, we employ a BiLSTM network with a CRF (Conditional Random Fields) layer to capture feature dependencies and generate an optimal global label sequence.

Assume the layout contains n markers, each with m dots. To standardize the input, dots outside the view are assigned a time step with features and labels set to 0, and valid time steps are padded accordingly, this can be represented as: $P_n = \{(d_{x_i}, d_{y_i})_{i \in [1, N]}, \dots, 0\} \in \mathbb{R}^{n \times m \times 2}$. The input of dots follows a nearest-neighbor-based ordering. For the input x and the corresponding classification y , the loss function is defined as:

$$P(y | x) = \frac{\exp(\text{score}(x, y))}{\sum_{y' \in \mathcal{Y}} \exp(\text{score}(x, y'))}. \quad (1)$$

$$\mathcal{L}(x, y) = -\log P(y | x), \quad (2)$$

Since the position of markers in world coordinate is known, if the correct pose can be obtained from successfully recognized dots, a RANSAC-based rematching algorithm can be applied to refine mismatched dots, thereby improving the accuracy of pose estimation.

3 RESULT

To evaluate the layout generation module, tests were conducted using a 9×7 grid involving 10 markers in one scenario and 14 markers in another, which is shown in Fig. 1 and Fig. 3. Set the number of dots per marker to 3 and cell size to $0.6m \times 0.6m$, we test the marker recognition module. The dataset excluded frames with less than six dots within the view and Gaussian noise with a mean of 0 and a variance of 1 is included. The data set encompasses 1 million poses randomly sampled within the tracking space. The training set, validation set, and test set are randomly partitioned in the ratio of 80%-10%-10%, the experiments have been performed on a PC with an NVIDIA RTX 3080 GPU, an Intel(R) Core(TM) i9-7900K processor. After validation, the number of BiLSTM layers is 3 and the units of BiLSTM are 256. The evaluation metrics used in this study are as follows: (1) The proportion of correctly recognized dots and the proportion of frames without error; (2) The average time taken to recognize a single frame. The results are presented in Tab. 1. The accuracy of the proposed model is higher than that of FC and CRF without correction algorithm (RE stands for the use of the rematch algorithm after the predicted result of CRF).

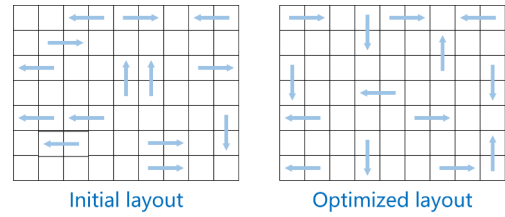


Figure 3: initial and optimized layout of grid 9×7 , 14 markers.

Table 1: Results using different models.

	(grid 9×7 , 10 markers)			(grid 9×7 , 14 markers)		
	FC	CRF	RE	FC	CRF	RE
Runtime/s	0.000014	0.000602	0.00132	0.000019	0.000678	0.00144
Dot acc	0.9894	0.9905	0.9928	0.9772	0.9838	0.9859
Frame acc	0.9585	0.9659	0.9812	0.9571	0.9651	0.9702

4 CONCLUSION & FUTURE WORK

This paper proposes a tracking system based on the recognition of positional features among sparse markers. By combining deep learning with a rematching algorithm, the rapid recognition of dots has been achieved with high accuracy.

Given the promising results in the virtual environment, this marks the first step towards practical applications, and we will develop a real tracking system by integrating an embedded camera with local PC processing. Additionally, network recognition can be easily affected by stray points, how to utilize the historical data to enhance recognition robustness is a promising research direction.

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